Literature Review

1 Introduction

1.1 Background

As higher education institutions face an increasingly diverse student body and the demand for more accessible and personalised education, chatbots emerge as an innovative solution to address a myriad of challenges [1].

When students move from high school to an unfamiliar university environment, they will naturally have many related questions to ask. Although university students are increasingly familiar with using the internet to obtain information, the information they need is often scattered across different websites, documents, and communication platforms. As a result, finding accurate and timely answers might be challenging.

In the paper by Yuwei Qian, it is pointed out that information-seeking behavior varies across individuals in different environments. This is especially true for freshmen, who may find it difficult to begin a university life that is very different from their high school experience [7]. This highlights the importance of providing structured and easily accessible information systems for freshmen.

Moreover, due to different living habits in different countries, students use a wide variety of online communication software. This also makes it difficult for universities to fully deliver daily-life information beyond official announcements.

Therefore, there is a need for a unified chatbot that can provide structured and reliable access to university information for freshmen. And in order to build the chatbot, a systematic review on previous chatbot applications is needed.

1.2 Purpose and Scope

The purpose of this literature review is to examine chatbot applications developed for university students, especially freshmen. It explores the previous applications of chatbots in the domain of education, the approaches adopted in these systems, and the evaluations metrics used for different types of chatbots, in order to better guide the development of the chatbot.

This review focuses on chatbot systems designed for higher education contexts, particularly those that support university students and freshmen through university enquiry services, student support, and FAQ-based assistance. It mainly examines traditional chatbot approaches, including rule-based, machine-learning-based, retrieval-based, and hybrid methods, while modern large language model (LLM)-based generative chatbots are discussed only briefly rather than in detail. Another focus of this review is the evaluation of chatbot effectiveness. Common evaluation aspects include classification performance, retrieval accuracy, response quality, and user satisfaction.

Together, these studies provide an overview of current developments in university chatbot systems and highlight areas that require further improvement.

1.3 Research Questions

RQ1. What existing chatbot applications have been developed to support university students?

RQ2. What techniques have been used in chatbot systems?

RQ3. What evaluation methods are used to assess diffenrent types of chatbots?

1.4 Search Strategy and Selection Criteria

The literature reviewed in this study was mainly collected through academic databases such as Google Scholar and IEEE Xplore. Sources include journal articles, conference papers, books, and peer-reviewed studies. The search focused on publications from 2008 to 2026, covering both early work on educational chatbots and more recent developments in AI-based conversational systems.

A combination of keyword searches and reference tracking was used to identify relevant studies. The main keywords included “college enquiry chatbot”, “university chatbot”, “student support chatbot”, “FAQ chatbot”, “rule-based chatbot”, “retrieval-based chatbot”, “hybrid chatbot”, and “information retrieval in chatbots”. These terms were selected to cover both the common applications of university chatbots and the different technical approaches discussed in this review. For several key papers, the reference lists were also examined to identify additional studies that were relevant to the topic.

During the selection process, attention was given not only to chatbot technologies but also to their use in higher education. Priority was therefore given to studies related to student support services, campus information enquiries, FAQ systems, and chatbot design for educational settings. Since this project focuses on a hybrid chatbot that combines rule-based matching, machine learning classification, and information retrieval, studies discussing these approaches were included as important references.

Studies focusing mainly on e-commerce, customer service, or other non-educational applications were not considered as primary sources. In addition, papers that only introduced chatbot concepts without discussing system implementation, evaluation methods, or practical applications were not examined in detail. The final selection of literature provides an overview of existing university chatbot applications, the technologies used to develop them, and the methods used to evaluate their effectiveness. These studies form the basis for addressing the research questions presented in this review.

2 Existing Chatbot Applications for University Students and Freshmen

2.1 University Enquiry Chatbots

University enquiry chatbots are primarily designed to improve access to campus-related information and streamline FAQ services. Their main role is to assist students in obtaining administrative information, announcements, and other university resources through a conversational interface. As a result, they represent one of the most common chatbot applications in higher education.

Susanna et al. developed a College Enquiry Chatbot that allows students to access information from links, text notifications, and PDF documents through a single platform [3]. Similarly, Salve et al. proposed a chatbot for university enquiries, but their system additionally incorporates a feedback mechanism that enables students to report inaccurate responses and allows administrators to refine the knowledge base accordingly [4].

Although both studies focus on improving information accessibility, they highlight different priorities. Susanna et al. emphasize the integration and delivery of information, whereas Salve et al. place greater importance on maintaining answer quality through continuous updates. This comparison suggests that effective enquiry chatbots require not only accessible information but also mechanisms to ensure the reliability and relevance of responses over time.

Despite their benefits, these systems largely depend on predefined knowledge bases and structured information sources. Their effectiveness is therefore closely tied to the quality and coverage of the available data. Nevertheless, the studies demonstrate that enquiry chatbots can play an important role in reducing information barriers and supporting students in navigating university services.

2.2 Student Support and FAQ Chatbots

Compared with enquiry chatbots, student support and FAQ chatbots address a broader range of student needs. Rather than focusing solely on information retrieval, these systems aim to support communication, learning, and student service management.

Dhandayuthapani proposed a student support chatbot framework that supports multilingual interaction and multiple communication channels, including text, voice, and graphical interfaces [5]. The study argues that such systems can improve service efficiency while reducing staff workload. Similarly, Nourbakhsh and Yaser discussed the wider educational role of chatbots, suggesting that they can support learning, strengthen communication between students and teachers, and improve access to educational opportunities [6].

Taken together, these studies indicate a gradual shift in higher education chatbots from information-centered systems towards more comprehensive student support platforms. Unlike enquiry chatbots, which mainly provide institutional information, student support chatbots aim to assist students throughout different aspects of their academic journey. This broader focus reflects the growing expectation that chatbots should contribute not only to information delivery but also to the overall student experience.

While the potential advantages are widely recognized, much of the literature remains focused on conceptual frameworks and expected outcomes. Consequently, further investigation is needed to understand how these systems perform in real university environments and how students perceive their usefulness in practice.

2.3 Limitations of Existing University Chatbot Applications

Although university chatbots have demonstrated significant potential, several limitations remain. Raval noted that chatbots may struggle to understand slang, informal expressions, or emotionally charged language, which can result in misunderstandings or repetitive response loops [9]. Such issues may reduce response accuracy and negatively affect user experience.

Another challenge is the highly institution-specific nature of university information. Since each university operates under different policies, procedures, and organizational structures, chatbot systems require extensive data collection and continuous maintenance. Ensuring that information remains accurate and up to date therefore represents an ongoing challenge.

Furthermore, many existing systems rely heavily on predefined knowledge bases and fixed response patterns. While this approach can effectively handle routine enquiries, it may be less effective when users ask ambiguous, unexpected, or complex questions. Existing studies also place greater emphasis on functionality and system design than on long-term user satisfaction or practical effectiveness.

Taken together, these limitations suggest that a single approach may not be sufficient to address the diverse needs of university students. This highlights the potential value of hybrid chatbot systems that combine rule-based methods, machine learning techniques, and information retrieval mechanisms to provide more flexible and reliable support.

3 Technical Approaches Used in University Chatbots

In the development of university FAQ chatbots, what to choose of technical architecture directly determines the overall performance of systems. To address the diverse needs of students, modern conversational agents have evolved from single method frameworks into systems integrated with multi-modular. This section evaluates various methods and deeply analyzes the core text classification algorithms supporting intent routing.

3.1 Rule-Based Approaches

Rule-based chatbots represent the earliest paradigm in educational systems. They provide specific responses relied on predefined rules and patterns for matching user inputs. Among the reviewed studies, we find Artificial Intelligence Markup Language (AIML) is commonly used to create rule-based chatbots. Khin and Soe [10], for example, developed a university chatbot using AIML to provide predefined responses to frequently asked questions. Since they are easy to build, developers tend to use them to address well-defined queries, such as assignments distribution or course schedules [11, 12, 13], which can greatly reduce the effort and cost to build other chatbots. However, Lopez and Qamber [14] observed that chatbots in higher education may struggle when users employ unexpected expressions or submit complex and ambiguous questions. Therefore, the reviewed literature suggests that their reliance on fixed patterns and short for semantic understanding makes them appropriate for narrowly defined queries, which are generally insufficient for university chatbot.

3.2 Machine-Learning-Based Approaches

To overcome the rigidity of pattern matching, researchers have increasingly used machine-learning-based (ML-based) methods for intent and category classification. As the development of natural language processing and machine learning algorithms, the methods for extracting text feature have been thriving. In the text classification task, the term frequency-inverse document frequency (TF-IDF) algorithm is widely used for its simplicity, strong interpretability, and suitability for small and medium-sized corpora. Its simple structure and clearing feature meaning make it adapt to constrained dialogue system in FAQ chatbots [15, 16]. Based on the text feature extraction, we evaluate three core text classification algorithms:

- **Naive Bayes:** Naive Bayes is a probabilistic machine learning algorithm that can be used in a wide variety of classification tasks. In early text classification, the multivariate Bernoulli model only recorded the presence or absence of a word, ignoring the term frequency. A study found that Multinomial Naive Bayes (MNB) model, which incorporates word frequency, performs better at larger vocabulary sizes [17]. However, an unseen word may produce a zero probability if it does not occur in the training data. To addressed this issue, Setyaningsih and Listiowarni [18] applied Laplace smoothing to ensure the robustness of inputs. And it is also proved to minimize misclassification errors. Consequently, MNB with smoothing provides a efficient and relatively robust solution for intent classification, especially when the available training dataset is limited.

- **Logistic Regression:** Among the papers we collected, Logistic Regression is another widely used method. Instead of fitting a joint probability distribution, it directly models the posterior probability, which maps from features to class labels. It can model the relationship between TF-IDF features and intent categories. However, Acito pointed out that some rare or unseen words may be assigned extreme weights, which can cause overfitting, misleading models in real world scenario[19]. To overcome this problem, many developers applied L2 regularization to improve generalizability and mitigate overfitting.

- **Linear SVM:** SVM is a geometric classification method based on structural risk minimization. It searches the best compromise between the complexity of the model (the learning accuracy of a particular training sample) and learning ability (error-free ability to identify any samples) in order to expect the best generalization ability [20]. An experiment shows that F1 value of SVM classifier has reached more than 86.26%, and the classification results comparing to other classification methods have greatly improved, proving that SVM is an effective machine learning method [21]. These findings predicate that Linear SVM is particularly suitable for high-dimensional and sparse textual features.

Overall, the reviewed studies demonstrate that MNB, Logistic Regression, and Linear SVM are all viable algorithms for intent classification. MNB offers computational simplicity and performs effectively with word-frequency features, Logistic Regression provides probabilistic outputs, and Linear SVM often achieves strong classification performance in high-dimensional feature spaces. However, the cited studies were conducted on different datasets and tasks. Their reported numerical results therefore cannot be compared directly. For the proposed chatbot, all three algorithms should be evaluated using the same university-query dataset.

3.3 Retrieval-Based Approaches

A retrieval-based chatbot works by matching user inputs to the most relevant responses within a predefined dataset. The responses here are entered manually by the developer, or based on a knowledge base of pre-existing information. Traditional systems commonly use lexical retrieval methods such as TF-IDF with cosine similarity or BM25 for their efficiency and interpretability [23, 24]. However, these methods may perform poorly when users express the same meaning with different vocabulary. In order to solve this limitation, researchers have introduced semantic retrieval methods based on BERT, Sentence-BERT, and dense passage representations, which can better identify paraphrases and semantically related questions [25, 26, 27]. Although these approaches improve retrieval flexibility, their effectiveness still depends heavily on the quality of the knowledge base. Therefore, retrieval-based methods are particularly suitable for university service chatbots because they provide reviewed and traceable answers to official enquiries.

3.4 Hybrid Chatbot Architectures

Hybrid Chatbot have been introduced to combine the rule-based systems, the ML models and retrieval-based or generative components. In a hybrid architecture, an intent classification model is first used to identify the type of user query and then direct it to the most appropriate module. Researchers such as Mikael et al. [22] proposed a hybrid chatbot for educational administration. They used a MNB classifier for query routing. Questions related to registration or financial matters were handled by the rule-based component, while academic and campus-related questions were directed to a retrieval-based module. For more complex or open-ended queries, they are forwarded to the generative model. By combining these different approaches, the system is able to improve both response reliability and conversational flexibility. It is also reported an overall accuracy of 97.57% in their experiments.

The reviewed hybrid studies show that different modules can be assigned according to their respective strengths. However, hybrid architectures are more difficult to design and maintain. An incorrect intent prediction may direct a query to an unsuitable module even when the required information is available elsewhere in the system. As a result, it is important to evaluate both classifier performance and end-to-end chatbot performance.

3.5 Comparison of Methods

To explicitly justify the technical selection for our project, Table 1 summarizes the strengths, limitations, and suitability of the discussed chatbot approaches.

In conclusion, each chatbot approach has its own strengths and limitations. To leverage the strengths of both approaches, our project adopts a hybrid architecture. Moreover, an MNB

Table 1: Comparison of chatbot approaches

Approach	**Strength**	**Limitation**	**Suitability for Our Project**
Rule-based	Controllable and simple	Poor flexibility	Good for greetings/help
ML-based	Handles varied wording	Needs labelled data	Good for category classification
Retrieval-based	Uses reviewed answers	Depends on FAQ coverage	Good for official FAQ answers
Generative	Flexible	Risk of hallucination	Not main method
Hybrid	Balances control and flexibility	More complex	Most suitable

classifier is used to identify user intent and route queries to the appropriate module. This design is expected to improve accuracy and usability in a university service environment.

4 Evaluation Methods for Chatbots

4.1 Evaluation Metrics for Machine Learning Category Classification

With the emergence of different types of chatbots, machine learning is widely used to allocate tasks to chatbots with different strengths. And evaluating the classification becomes a need. Luckily, machine learning is quite a mature technique, and most of the classic evaluation metrics for machine learning can be directly applied here. In the 34 papers we collected, 32 used classification accuracy to measure, while 30 of them calculated the F1-score for their classification models. Precision and recall are also widely used, with both 29 papers reporting them. Moreover, the confusion matrix, ROC-AUC, task success rate are also applied to evaluate the model performance, although they were less common.

4.2 Evaluation Metrics for Rule-Based Chatbots

Since around 2018–2020, research shifted heavily towards retrieval-based chatbots, generative chatbots, and LLM-based agents, there are quite few papers talking about the evaluation metrics for pure rule-based chatbots. In the eight papers we collected, rule-based chatbots are generally evaluated from four aspects: Coverage, Robustness, User satisfaction.

Coverage Coverage is used to measure how much of the users' question space this chatbot can handle. An ideal chatbot should be able to response to most of the users' questions. And that's the reason why coverage should be considered as one of the core measurements for a chatbot's capability. To measure this, D'Ávila [29] introduced the KINO framework, which evaluates rule-based chatbot performance using two monitoring metrics—assertiveness (the ability to provide a valid response) and ambiguity (the degree of uncertainty caused by overlapping rules).

Robustness Sometimes, users might click Enter accidentally without finishing typing their questions, or they might mistaken the spelling of some words. Therefore, robustness should also be considered as one of the important standards to evaluate a chatbot. D'Ávila [29] used out-of-scope (OOS) interactions in his KINO framework to refer to user inputs that cannot be matched to any existing conversational rule, intent, or response pattern. A high proportion of OOS inputs suggests reduced conversational robustness, whereas a lower OOS rate indicates that the chatbot can successfully handle a broader range of user requests within its intended domain.

User satisfaction As a chatbot designed to provide users with convenience, user satisfaction is absolutely an important standard to evaluate a chatbot. To measure this, Papakostas et al. [30] used a questionnaire administered to 50 undergraduate students. The questionnaire covered general user satisfaction, query accuracy, response time, and personalized guidance. By directly questioning users, the authors can get a grasp of the actual user satisfaction conveniently. However the problem is, surveying a sufficiently large number of people might be a costly and labor-intensive work, whereas surveying just a small number of respondents may lead to biased conclusion. Moreover, Dhinagaran et al. [31] provided another way to evaluate user satisfaction. They use retention (the proportion of users who continue using the chatbot over time) and number of interactions (the number of rounds a user need before their query is resolved) to measure the user satisfaction. If a user finds the chatbot helpful, they will keep using it, and if the chatbot can solve the user's problem efficiently, the user will get satisfied. Therefore, these two are indeed reasonable evaluation metrics.

4.3 Evaluation Metrics for Retrieval-Based Chatbots

For retrieval-based chatbots, Recall@k (measures whether the correct answer appears within the top k retrieved responses), MRR (measures how highly the first correct answer is ranked), and Precision@k (measures how many of the top k retrieved results are actually relevant) are most commonly used. Metrics like ROUGE and BERTScore are also frequently mentioned, but they are more commonly used for retrieval-augmented generation (RAG) chatbots rather than pure retrieval-based chatbots, so they are not the focus here.

Interestingly, another paper we found pointed out the limitation of the previously mentioned metrics. Liu et al. [32] performed a meta-evaluation. Instead of evaluating chatbots, they evaluate the classic evaluation metrics themselves. Their finding indicates that most commonly used metrics do not strongly predict what users actually prefer. A model with high Recall@k, Precision@k, etc. can still provide poor user experience. Inspired by their findings, we propose that a complete evaluation for retrieval based chatbots should consist of two complementary literature streams: Response Selection Evaluation (classic evaluation) and Conversational Evaluation Metrics (user satisfaction). In this way the developer can have a comprehensive grasp of the model performance.

5 Conclusion

This literature review examined existing university chatbot applications, the technical approaches used in chatbot development, and the evaluation methods adopted in previous studies.

The review found that university chatbots are widely applied in areas such as student enquiries, academic support, admissions, and administrative services. And the chatbots can indeed help freshmen adapt to university life more quickly. Among the different approaches, they each has their own strengths and limitation. Recent studies suggest that hybrid architectures might combine their advantages and improve the chatbot performance. The review also analyzed the commonly used evaluation metrics for diffenrent chatbot components. And concluded that a comprehensive assessment should be composed of evaluation from different aspects.

Overall, this literature review analyses the chatbots built by previous researchers, and can be a good starting point to build FAQ chatbots.

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